



US012403853B2

(12) **United States Patent**
Mukaigawa et al.

(10) **Patent No.:** **US 12,403,853 B2**

(45) **Date of Patent:** **Sep. 2, 2025**

(54) **VEHICLE**

(71) Applicant: **TOYOTA JIDOSHA KABUSHIKI**
KAISHA, Toyota (JP)

(72) Inventors: **Yuuki Mukaigawa**, Seto (JP); **Naoki Tani**, Nagoya (JP); **Ryuma Osaka**, Toyota (JP); **Shungo Iizuka**, Toyota (JP); **Shunji Shibata**, Anjo (JP); **Kyohei Miyaoka**, Ikeda (JP); **Masanori Koderu**, Toyota (JP); **Takashi Iwasaki**, Hamamatsu (JP)

(73) Assignee: **TOYOTA JIDOSHA KABUSHIKI**
KAISHA, Toyota (JP)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 318 days.

(21) Appl. No.: **18/126,708**

(22) Filed: **Mar. 27, 2023**

(65) **Prior Publication Data**

US 2023/0311795 A1 Oct. 5, 2023

(30) **Foreign Application Priority Data**

Mar. 29, 2022 (JP) 2022-053204

(51) **Int. Cl.**

B60R 16/04 (2006.01)

B60K 1/04 (2019.01)

B62D 21/15 (2006.01)

B62D 25/20 (2006.01)

(52) **U.S. Cl.**

CPC **B60R 16/04** (2013.01); **B60K 1/04** (2013.01); **B62D 21/155** (2013.01); **B62D 25/20** (2013.01); **B60K 2001/0438** (2013.01)

(58) **Field of Classification Search**

CPC B60K 1/04; B60K 2001/0438; B62D 21/155; B62D 21/157

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

9,809,101 B2 *	11/2017	Ikeda		B62D 29/001
10,710,638 B2 *	7/2020	Kawase		B60K 1/04
2014/0117716 A1 *	5/2014	Patberg		B62D 21/15
				296/187.08
2015/0176673 A1	6/2015	Kaneko et al.		
2016/0083017 A1	3/2016	Sakaguchi et al.		
2016/0114667 A1 *	4/2016	Ikeda		H01M 50/24
				180/68.5

(Continued)

FOREIGN PATENT DOCUMENTS

JP	2008-174181 A	7/2008
JP	2013-256265 A	12/2013

(Continued)

Primary Examiner — Brian L Swenson

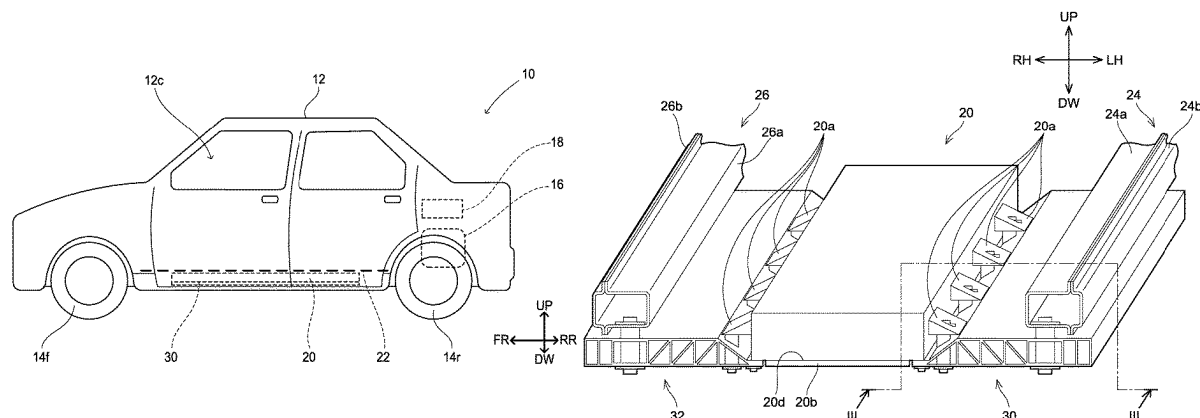
(74) *Attorney, Agent, or Firm* — Dinsmore & Shohl LLP

(57)

ABSTRACT

A vehicle may include a vehicle body including a floor panel, a battery pack located below the floor panel and an energy absorption member located outwardly from the battery pack in a vehicle width direction and fixing the battery pack to the vehicle body. A support that protrudes downwardly may be disposed on a lower wall of the battery pack. The energy absorption member may include a protrusion that protrudes inwardly in the vehicle width direction toward the support. A tip end of the protrusion may include a contact surface being in contact with a side surface of the support in the vehicle width direction.

10 Claims, 7 Drawing Sheets



(56)

References Cited

U.S. PATENT DOCUMENTS

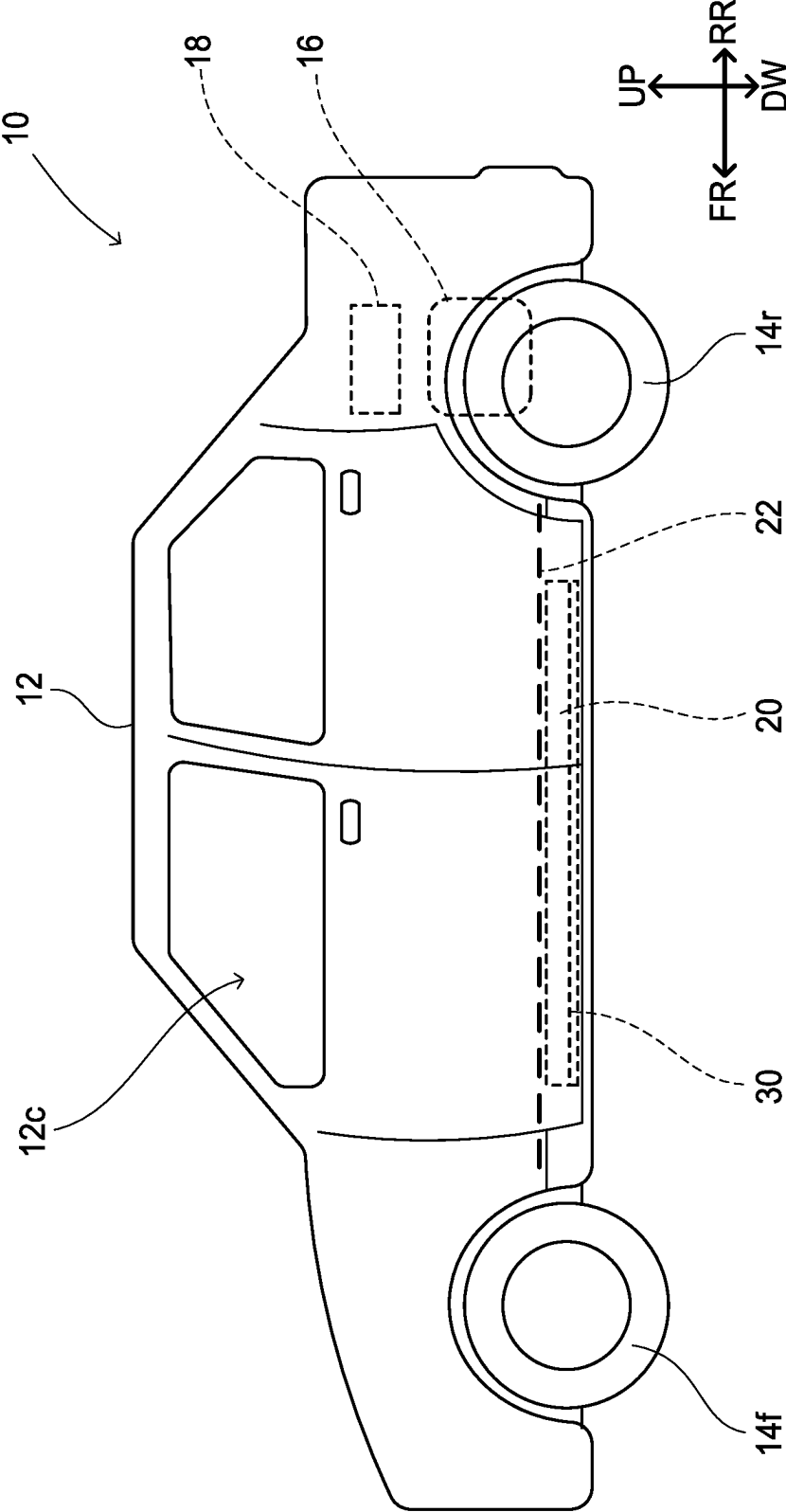
2018/0072352	A1 *	3/2018	Shimoda	B60K 6/28
2021/0221435	A1	7/2021	Takahashi et al.	
2022/0185379	A1 *	6/2022	Kim	B60K 1/04
2022/0227215	A1 *	7/2022	Ishizaki	B62D 21/157
2023/0202302	A1 *	6/2023	Jo	B62D 21/11
				180/68.5
2024/0025486	A1 *	1/2024	Watanabe	B60R 3/02

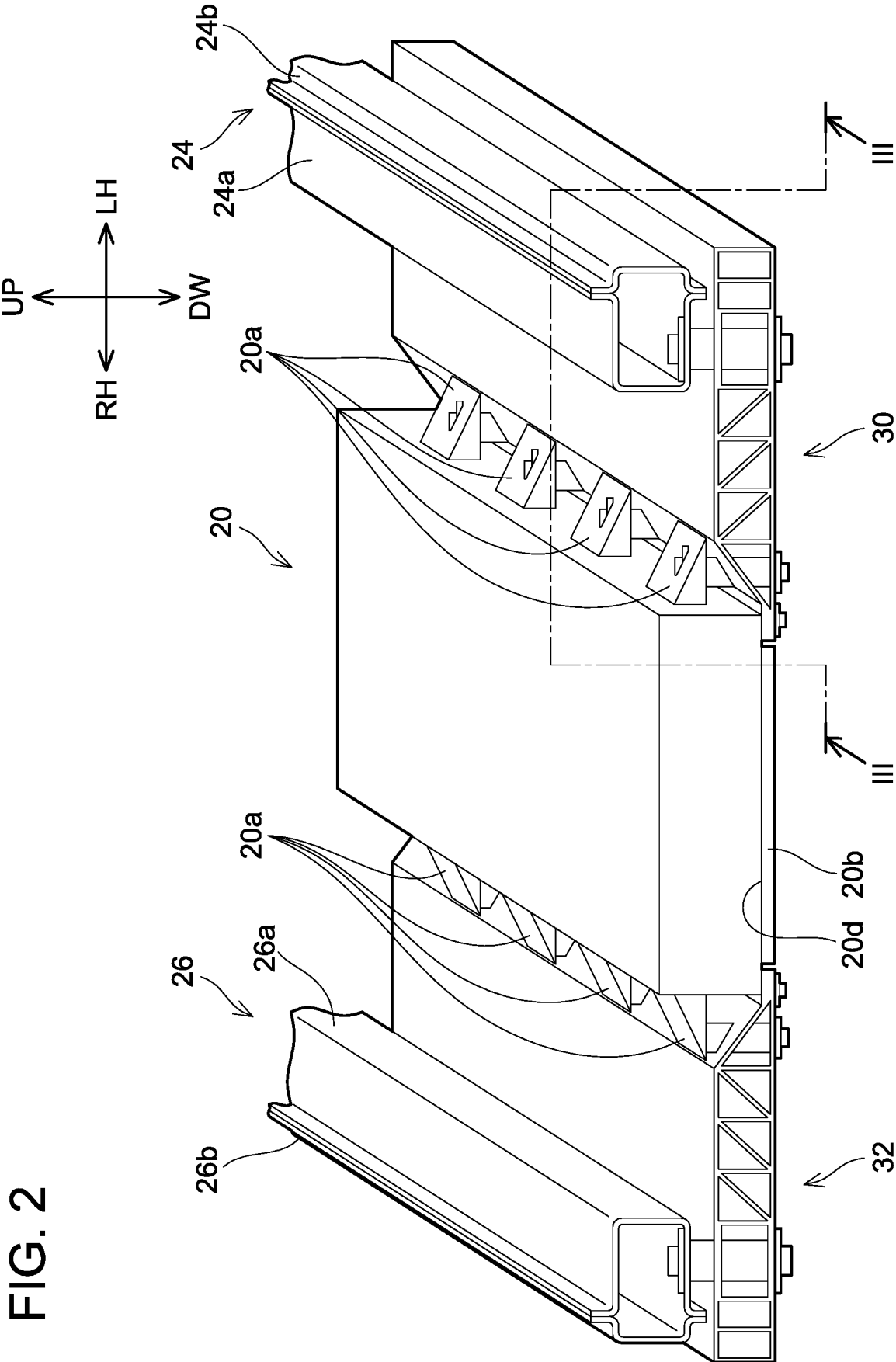
FOREIGN PATENT DOCUMENTS

JP	2014-031109	A	2/2014
JP	2014-080116	A	5/2014
JP	2014-227075	A	12/2014
JP	2017-226353	A	12/2017
JP	2021-112973	A	8/2021

* cited by examiner

FIG. 1





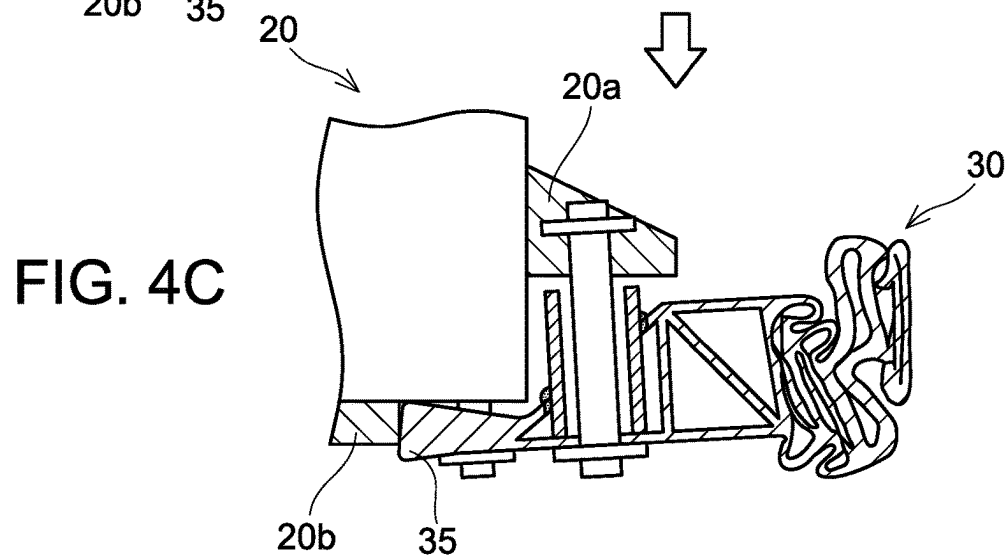
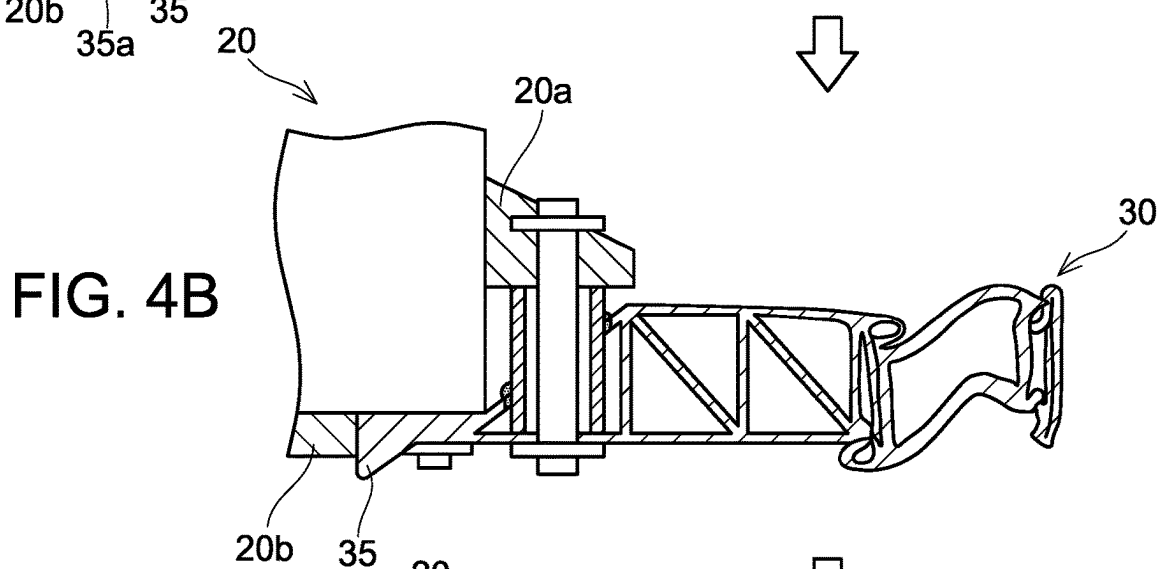
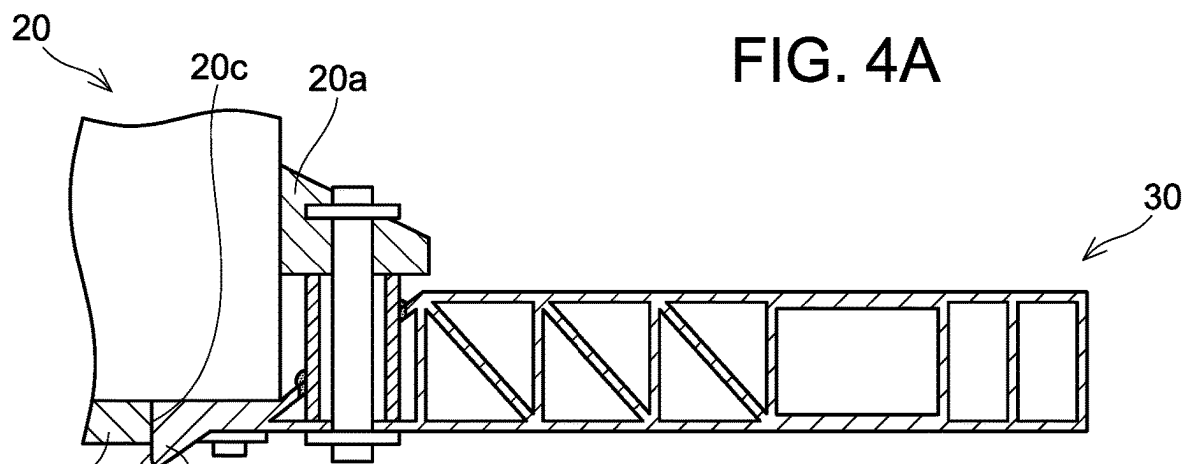


FIG. 5

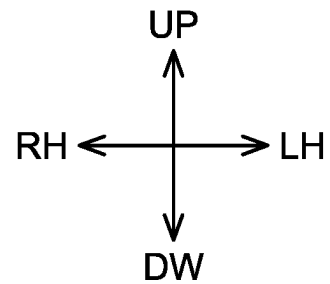
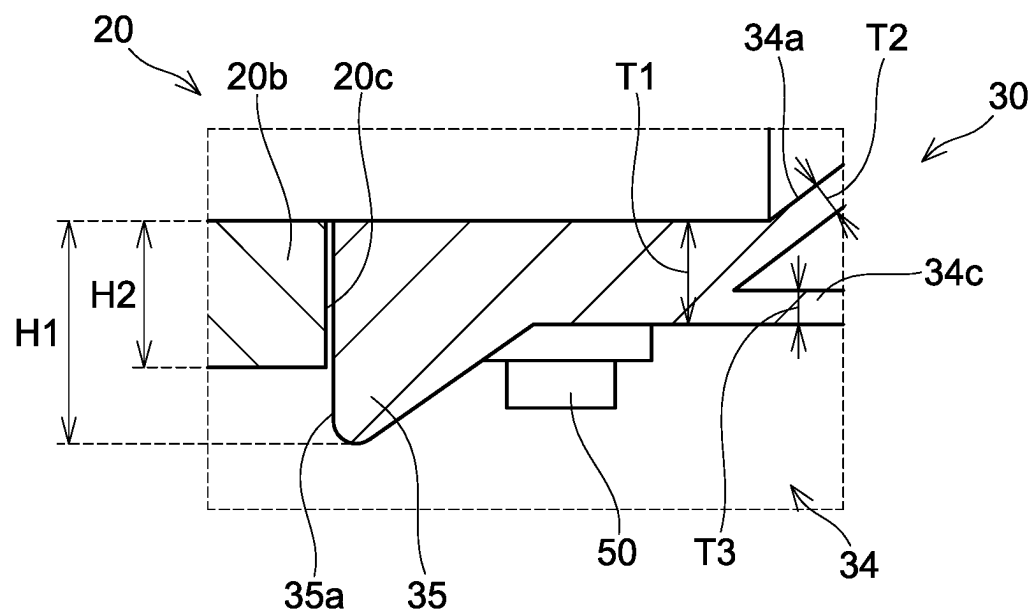


FIG. 6

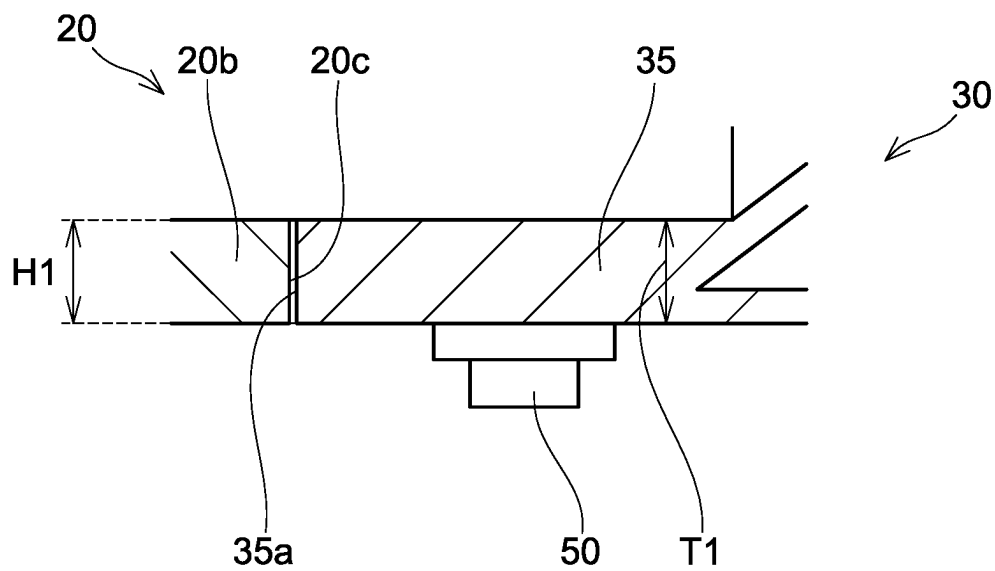


FIG. 7

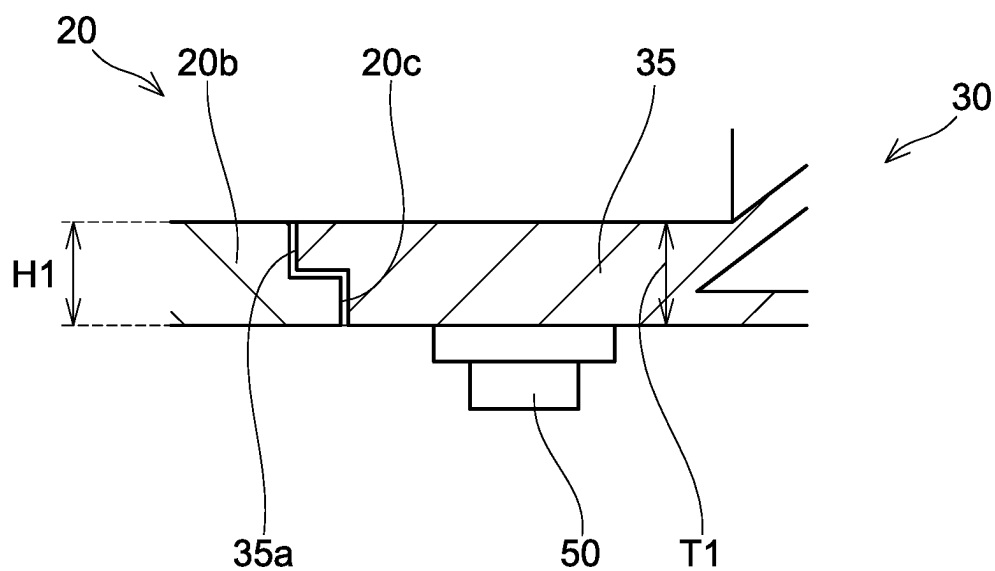


FIG. 8

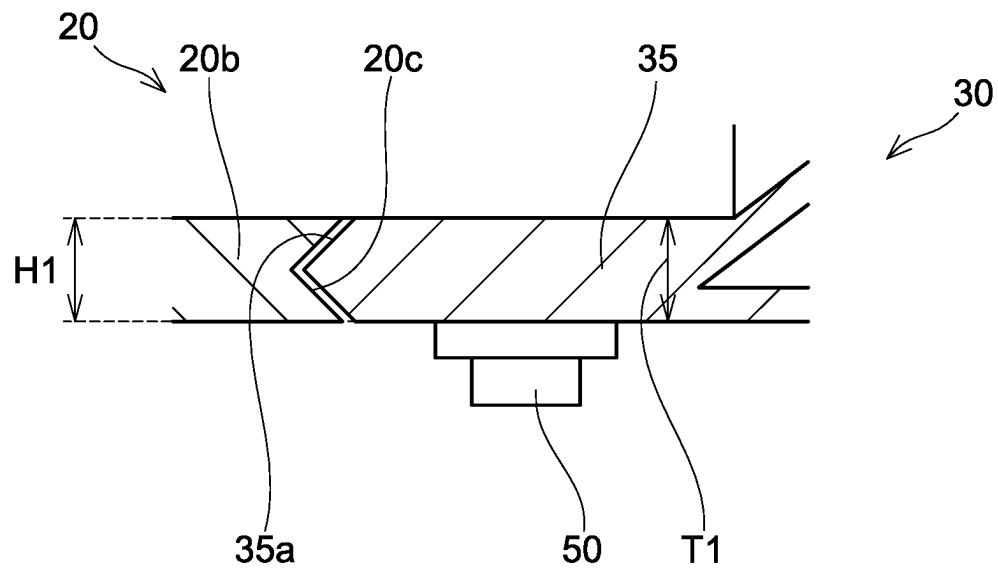
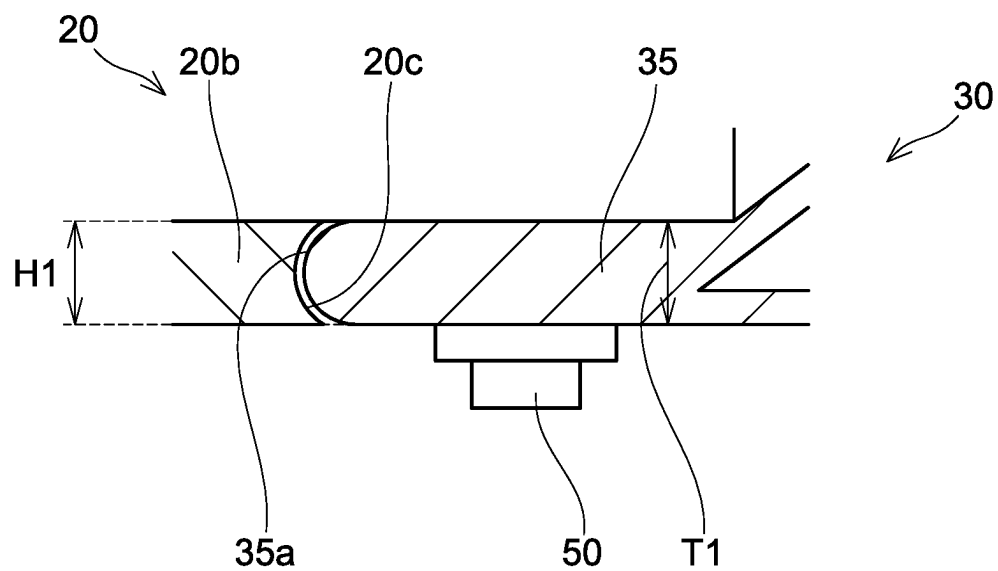


FIG. 9



1

VEHICLE

CROSS-REFERENCE TO RELATED APPLICATION(S)

This application claims priority to Japanese patent application No. 2022-053204, filed on Mar. 29, 2022, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

The technique disclosed herein relates to a vehicle.

BACKGROUND

Japanese Patent Application Publication No. 2017-226353 describes a vehicle. This vehicle includes a pair of rockers, a floor panel extending between the pair of rockers, a battery pack located below the floor panel, and an energy absorption member located outwardly from the battery pack in a vehicle width direction and fixed to one of the pair of rockers.

SUMMARY

With the above structure, when side collision occurs to the vehicle, an energy absorption member (which hereafter may be referred to as an EA member) deforms, by which collision energy is absorbed. At this time, if an inner sidewall of the EA member facing the battery pack is pressed against a sidewall of the battery pack, relatively large collision load may be applied on the sidewall of the battery pack.

In light of the above situation, the present disclosure provides a technique which may reduce collision load transmitted from an EA member to a sidewall of a battery pack.

The technique disclosed herein is implemented as a vehicle. This vehicle may comprise: a vehicle body comprising a floor panel; a battery pack located below the floor panel; and an energy absorption member located outwardly from the battery pack in a vehicle width direction and fixing the battery pack to the vehicle body. A support that protrudes downwardly may be disposed on a lower wall of the battery pack. The energy absorption member may comprise a protrusion that protrudes inwardly in the vehicle width direction toward the support. A tip end of the protrusion may comprise a contact surface being in contact with a side surface of the support in the vehicle width direction.

With the above structure, when side collision occurs to the vehicle, the EA member deforms, by which collision energy is absorbed. The EA member comprises the protrusion that protrudes inwardly in the vehicle width direction toward the support disposed on the lower wall of the battery pack, and the tip end of the protrusion comprises the contact surface being in contact with the side surface of the support. Thus, when side collision occurs to the vehicle, collision load applied on the EA member is transmitted to the support disposed on the lower wall of the battery pack via the protrusion. Consequently, collision load transmitted to the sidewall of the battery pack from the EA member can be reduced.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 schematically illustrates a configuration of a vehicle 10 of Embodiment 1.

2

FIG. 2 is a view explaining positional relationships between a battery pack 20, a pair of EA members 30, 32 and a pair of rockers 24, 26.

FIG. 3 is a cross-sectional view along a line of FIG. 2.

FIG. 4A illustrates an example of deformation of the EA member 30 at time when side collision occurs to the vehicle 10; FIG. 4B illustrates an example of deformation of the EA member at a subsequent time when side collision occurs to the vehicle; and FIG. 4C illustrates an example of deformation of the EA member at another subsequent time when side collision occurs to the vehicle.

FIG. 5 is an enlarged view of an enclosed area V in FIG. 3.

FIG. 6 is a view to explain a variant of shapes of a contact surface 35a of a protrusion 35 and a side surface 20c of a support 20b.

FIG. 7 is a view to explain a variant of shapes of the contact surface 35a of the protrusion 35 and the side surface 20c of the support 20b.

FIG. 8 is a view to explain a variant of shapes of the contact surface 35a of the protrusion 35 and the side surface 20c of the support 20b.

FIG. 9 is a view to explain a variant of shapes of the contact surface 35a of the protrusion 35 and the side surface 20c of the support 20b.

DETAILED DESCRIPTION

In an embodiment of the technique disclosed herein, a dimension of the contact surface of the protrusion in a vehicle height direction may be equal to or greater than a dimension of the side surface of the support in the vehicle height direction. According to such a configuration, when side collision occurs to the vehicle, it is possible to avoid or suppress the protrusion of the EA member being locally pressed against the side surface of the support disposed on the battery pack. Consequently, it is possible to avoid or suppress collision load transmitted to the support being locally concentrated.

Alternatively to the aforementioned embodiment, a dimension of the contact surface of the protrusion in a vehicle height direction may be greater than a dimension of the side surface of the support in the vehicle height direction. As another embodiment, however, the dimension of the contact surface of the protrusion in the vehicle height direction may be equal to or smaller than the dimension of the side surface of the support in the vehicle height direction.

In an embodiment of the technique disclosed herein, at least a part of the protrusion may be located below the battery pack and extend along the lower wall of the battery pack. According to such a configuration, the dimension of a portion of the EA member in the vehicle width direction which is advantageous to absorb collision energy can be increased by the dimension of the protrusion in the vehicle width direction located below the battery pack. Consequently, when side collision occurs to the vehicle, the EA member deforms, by which collision energy to be absorbed can be increased.

In the aforementioned embodiment, the protrusion may be fixed to the lower wall of the battery pack with a bolt. According to such a configuration, the protrusion can firmly be fixed to the battery pack. Thus, when the EA member is compressed and deformed in the vehicle width direction toward to the sidewall of the protrusion, it is possible to avoid or suppress the protrusion being unintentionally separated away from the battery pack. As another embodiment,

however, the protrusion may be fixed to the lower wall of the battery pack with, for example, an adhesive instead of the bolt.

In an embodiment of the technique disclosed herein, the protrusion may have a plate-like shape extending in the vehicle width direction. In this case, a thickness of the protrusion may be greater than a thickness of any one of walls that constitutes a rest of the energy absorption member. According to such a configuration, it is possible to make rigidity of the protrusion having the plate-like shape greater than rigidity of any one of the walls that constitute the rest of the EA member except the protrusion. Consequently, it is possible to make the protrusion less susceptible to deformation as compared to any one of the walls that constitute the rest of the EA member when the EA member is compressed and deformed in the vehicle width direction toward the battery pack. Thus, the collision energy generated in side collision occurring to the vehicle is primarily absorbed by deformation of a part of or an entirety of the rest of the EA member except the protrusion.

In an embodiment of the technique disclosed herein, in at least a part of the protrusion adjacent to the contact surface, a thickness of the protrusion may increase toward the contact surface. According to such a configuration, it is possible to make a minimum ground clearance of the protrusion (i.e., a distance from the ground surface to the lowest surface of the protrusion in the vehicle height direction) relatively great while rigidity of the contact surface of the protrusion is maintained.

In an embodiment of the technique disclosed herein, the support may be formed on a member that is attached to the battery pack. As another embodiment, however, the support may be formed integrally with the battery pack.

In an embodiment of the technique disclosed herein, the contact surface of the protrusion and the side surface of the support may have uneven shapes that engage each other. According to such a configuration, it is possible to increase an area where the contact surface of the protrusion is in contact with the side surface of the support. In particular, when the uneven shapes are defined along the vehicle height direction, it is possible to suppress the contact surface of the protrusion moving away from the side surface of the support when side collision occurs to the vehicle. Consequently, when side collision occurs to the vehicle, collision load applied on the EA member is effectively transmitted to the support disposed on the lower wall of the battery pack via the protrusion.

The simple recitations such as “forward”, “rearward”, “a longitudinal direction” herein respectively refer to forward, rearward and a longitudinal direction with respect to a vehicle. Similarly, the simple recitations such as “leftward”, “rightward” and “a vehicle width direction” respectively refer to leftward, rightward and a vehicle width direction with respect to the vehicle and the simple recitations such as “upward”, “downward” and “a height direction” refer to upward, downward and a height direction with respect to the vehicle. The vehicle width direction of the vehicle is also a left-right direction of the vehicle, and it may be referred to as “left-right direction” herein. For example, when the vehicle is placed on a horizontal surface, the height direction of the vehicle coincides with the vertical direction. The vehicle width direction of the vehicle is a direction parallel to the horizontal surface and parallel to an axle of the vehicle, and the longitudinal direction of the vehicle is a direction parallel to the horizontal surface and perpendicular to the axle of the vehicle.

Representative, non-limiting examples of the present disclosure will now be described in further detail with reference to the attached drawings. This detailed description is merely intended to teach a person of skill in the art further details for practicing aspects of the present teachings and is not intended to limit the scope of the present disclosure. Furthermore, each of the additional features and teachings disclosed below may be utilized separately or in conjunction with other features and teachings to provide improved vehicles, as well as methods for using and manufacturing the same.

Moreover, combinations of features and steps disclosed in the following detailed description may not be necessary to practice the present disclosure in the broadest sense, and are instead taught merely to particularly describe representative examples of the present disclosure. Furthermore, various features of the above-described and below-described representative examples, as well as the various independent and dependent claims, may be combined in ways that are not specifically and explicitly enumerated in order to provide additional useful embodiments of the present teachings.

All features disclosed in the description and/or the claims are intended to be disclosed separately and independently from each other for the purpose of original written disclosure, as well as for the purpose of restricting the claimed subject matter, independent of the compositions of the features in the embodiments and/or the claims. In addition, all value ranges or indications of groups of entities are intended to disclose every possible intermediate value or intermediate entity for the purpose of original written disclosure, as well as for the purpose of restricting the claimed subject matter.

EMBODIMENT

With reference to drawings, a vehicle **10** of an embodiment will be described. The vehicle **10** of the present embodiment is classified as an electric vehicle including a motor **16** driving wheels **14f**, **14r**, and is typically an electric vehicle which travels on roads (a so-called automobile). However, a part of or all the technique described in the present embodiment can similarly be applied to an electric vehicle which travels on tracks. The vehicle **10** is not limited to a vehicle operated and driven by a user, but may be a vehicle remotely operated by external device(s) or a vehicle which travels autonomously.

Here, a direction FR in the drawings indicates a front side in a longitudinal direction (or a front-rear direction) of the vehicle **10**, and a direction RR indicates a rear side in the longitudinal direction of the vehicle **10**. A direction LH indicates a left side in a vehicle width direction (or a left-right direction) of the vehicle **10** and a direction RH indicates a right side in the vehicle width direction of the vehicle **10**. A direction UP indicates an upside in a height direction (or an up-down direction) of the vehicle **10**, and a direction DW indicates a downside in the height direction of the vehicle **10**.

As illustrated in FIG. 1, the vehicle **10** includes a vehicle body **12** and the plurality of wheels **14f**, **14r**. The vehicle body **12** includes a cabin **12c** which is a space to accommodate occupants. The plurality of wheels **14f**, **14r** is rotatably attached to the vehicle body **12**. The plurality of wheels **14f**, **14r** includes a pair of front wheels **14f** positioned at a front portion of the vehicle body **12** and a pair of rear wheels **14r** positioned at a rear portion of the vehicle body **12**. The pair of front wheels **14f** is disposed coaxially to each other, and the pair of rear wheels **14r** is also disposed

5

coaxially to each other. The number of wheels **14f**, **14r** is not limited to four. The vehicle body **12** is constituted of, but not particularly limited to, metal, such as steel or aluminum alloy.

As illustrated in FIG. 1, the vehicle **10** further includes a motor **16**, an electric control unit **18** and a battery pack **20**. The motor **16** is a traction motor that drives the pair of rear wheels **14r** and is connected to the pair of rear wheels **14r**. The battery pack **20** is an electric device configured to supply power to the motor **16** and is electrically connected to the motor **16** via the electric control unit **18**. The battery pack **20** houses a plurality of secondary battery cells and is rechargeable with external power or regenerative power of the motor **16**. The battery pack **20** is located below a floor panel **22** and is disposed along the floor panel **22**. Although this is merely an example, as illustrated in FIGS. 2 and 3, the battery pack **20** of the present embodiment includes brackets **20a** protruding outwardly in the vehicle width direction. A support **20b** that protrudes downwardly is disposed on a lower wall **20d** of the battery pack **20**. The support **20b** is for example a lower cover to protect the lower wall **20d** of the battery pack **20**, and is attached to the lower wall **20d** of the battery pack **20** as a member separate from the battery pack **20**. The electric control unit **18** houses a DC-DC converter and/or inverter and is configured to control driving power supplied from the battery pack **20** to the motor **16** and regenerative power supplied from the motor **16** to the battery pack **20** depending on, for example, driving operations by the user. It should be noted that it is not necessary for the support **20b** to be formed on a member that is attached to the battery pack **20**, and the support **20b** may integrally be formed with the battery pack **20**.

Wheel(s) to be driven by the motor **16** are not limited to the pair of rear wheels **14r**, and the motor **16** simply needs to be configured to drive at least one of the plurality of wheels **14f**, **14r**. The vehicle **10** may further include another prime mover such as an engine instead of or as an alternative to the motor **16**. The vehicle **10** may include another power device such as fuel cell unit(s) or solar panel(s) instead of or as an alternative to the battery pack **20**. The vehicle **10** is not limited to an electric automobile described herein but may be a hybrid vehicle, fuel-cell vehicle, a solar car, or the like.

As illustrated in FIGS. 2 and 3, the vehicle body **12** includes the floor panel **22**, a pair of rockers **24**, **26** and a floor crossbeam **28**. The floor panel **22** is a plate-like member constituting a bottom surface of the cabin **12c**. Each of the rockers **24**, **26** is a hollow member having a constant cross-section along the vehicle longitudinal direction, and constitutes a frame of the vehicle body **12**. The pair of rockers **24**, **26** includes a left rocker **24** and a right rocker **26**. The left rocker **24** is positioned at a left edge **22a** of the floor panel **22** and extends in the front-rear direction on the outer side of the floor panel **22** in the vehicle width direction. The pair of rockers **24** is arranged symmetrically to each other with respect to the vehicle width direction. Thus, the right rocker **26** is positioned at a right edge of the floor panel **22** and extends in the front-rear direction on the outer side of the floor panel **22** in the vehicle width direction.

Although not particularly limited to this configuration, the left rocker **24** includes a left rocker inner panel **24a** positioned on the inner side in the width direction and a left rocker outer panel **24b** positioned on the outer side in the width direction. The left rocker inner panel **24a** and the left rocker outer panel **24b** are joined to each other at the respective top and bottom edges, and a closed space extending in the front-rear direction is defined inside the left rocker **24**. Although not particularly limited to this configuration as

6

well, the right rocker **26** includes a right rocker inner panel **26a** positioned on the inner side in the width direction and a right rocker outer panel **26b** positioned on the outer side in the width direction. The right rocker inner panel **26a** and the right rocker outer panel **26b** are joined to each other at the respective top and bottom edges, and a closed space extending in the front-rear direction is defined inside the right rocker **26**. The floor panel **22** extends between the pair of rockers **24**, **26** and is joined to the left rocker inner panel **24a** at its left edge **22a** and joined to the right rocker inner panel **26a** at its right edge. The rockers **24**, **26** may be constituted of three or more panels, not limited only to the rocker inner panels **24a**, **26a** and the rocker outer panels **24b**, **26b**.

The floor crossbeam **28** is a hollow member having a constant cross section along the vehicle longitudinal direction and constitutes a part of the frame of the vehicle body **12**. The floor crossbeam **28** is positioned on the floor panel **22** and extends between the pair of rockers **24**, **26** in the vehicle width direction. Although the illustration is omitted, a plurality of floor crossbeams **28** is disposed between the pair of rockers **24**, **26**.

As illustrated in FIGS. 2, 3, the vehicle **10** further includes a pair of energy absorption members **30**, **32** (hereafter referred to as EA members **30**, **32**). Each of the pair of EA members **30**, **32** is a hollow member extending along the vehicle longitudinal direction and having a constant cross section along the vehicle longitudinal direction. The pair of EA members **30**, **32** includes a left EA member **30** and a right EA member **32**. The pair of EA members **30**, **32** is arranged symmetrically to each other with respect to the vehicle width direction. The left EA member **30** is located at a left portion of the vehicle **10** and outwardly from the battery pack **20** in the vehicle width direction. The right EA member **32** is located at a right portion of the vehicle **10** and outwardly from the battery pack **20** in the vehicle width direction. One of the EA members **30**, **32** is compressed and deformed when side collision occurs to the vehicle **10**, by which it absorbs collision energy. The pair of EA members **30**, **32** is constituted of metal such as aluminum. However, a material constituting the pair of EA members **30**, **32** is not particularly limited.

As described above, the pair of EA members **30**, **32** is arranged symmetrically to each other with respect to the vehicle width direction. Structures of the left EA member **30** and its relevant members (e.g., the battery pack **20** and the left rocker **24**) are symmetrical to structures of the right EA member **32** and its relevant members (e.g., the battery pack **20** and the right rocker **26**) with respect to the vehicle width direction. Thus, with reference to FIG. 3, the structures of the left EA member **30** and its relevant members (e.g., the battery pack **20** and the left rocker **24**) will hereafter be described. Hereafter, the left EA member **30** will simply be referred to as the EA member **30**.

As illustrated in FIG. 3, the EA member **30** includes an inner end portion **34**, a first middle portion **36**, a second middle portion **38** and an outer end portion **40**. The inner end portion **34** is at the innermost position in the vehicle width direction and faces the battery pack **20**. The outer end portion **40** is at the outermost position in the vehicle width direction. The first middle portion **36** and the second middle portion **38** are positioned between the inner end portion **34** and the outer end portion **40** in the vehicle width direction. The first middle portion **36** is adjacent to the inner end portion **34** and the second middle portion **38** is adjacent to the outer end portion **40**. In other words, the inner end portion **34**, the first middle portion **36**, the second middle

portion 38 and the outer end portion 40 are arranged in this order from inside to outside in the vehicle width direction.

Although not particularly limited to this configuration, the inner end portion 34 includes an inner sidewall 34a, a vertical wall 34b and a lower wall 34c. The inner sidewall 34a faces the battery pack 20 and inclines downwardly in the vehicle height direction as it extends inwardly in the vehicle width direction. The vertical wall 34b extends along the vehicle height direction and the lower wall 34c extends along the vehicle width direction. Thus, a space having a constant cross-section in the vehicle longitudinal direction is defined by the inner sidewall 34a, the vertical wall 34b and the lower wall 34c.

As illustrated in FIG. 3, the vehicle 10 further includes first collars 42 and first bolts 44. The inner sidewall 34a included in the inner end portion 34 of the EA member 30 comprises openings through which the first collars 42 extend. The first collars 42 are disposed such that the first collars 42 extend through the openings in the inner sidewall 34a in the vehicle height direction. Although this is an example, the first collars 42 are welded to the inner sidewall 34a of the EA member 30. The upper end of each first collar 42 is in contact with corresponding one of the brackets 20a of the battery pack 20 and the lower end of each first collar 42 is in contact with the lower wall 34c of the inner end portion 34. The first bolts 44 extend through the first collars 42 and fasten the EA member 30 to the brackets 20a of the battery pack 20. Thus, the EA member is fixed to the brackets 20a of the battery pack 20 at the inner end portion 34. An inner diameter D1 of the first collars 42 is greater than a nominal diameter D2 of the first bolts 44. For example, the inner diameter D1 of the first collars 42 is at least twice the nominal diameter D2 of the first bolts 44. As another embodiment, the brackets 20a may not be disposed on the battery pack 20, and the EA member 30 may directly be fastened to the battery pack 20 by the first bolts 44.

As illustrated in FIG. 3, the vehicle 10 further includes second collars 46 and second bolts 48. Although not particularly limited to this configuration, an upper wall 38a of the second middle portion 38 of the EA member 30 also includes openings through which the second collars 46 extend. The second collars 46 are disposed such that the second collars 46 extend through the openings in the upper wall 38a of the second middle portion 38 in the vehicle height direction. Although this is an example, the second collars 46 are welded to the upper wall 38a of the second middle portion 38. The upper ends of the second collars 46 are in contact with the left rocker inner panel 24a of the left rocker 24 and the lower ends of the second collars 46 are in contact with the lower wall 38b of the second middle portion 38. The second bolts 48 extend through the second collars 46 and fasten the EA member 30 to the left rocker 24. Thus, the EA member 30 is fixed to the left rocker 24 of the vehicle body 12 at the second middle portion 38. As described above, the EA member 30 is fixed to the brackets 20a of the battery pack 20 at the inner end portion 34, thus the EA member 30 fixes the battery pack 20 to the vehicle body 12. Although not particularly limited to this configuration, the inner diameter of the second collars 46 is greater than the nominal diameter of the second bolts 48.

It should be noted that, the position where the EA member 30 is fixed to the bracket 20a of the battery pack 20 does not need to be the inner end portion 34. The position where the EA member 30 is fixed to the bracket 20a of the battery pack 20 can suitably be changed depending on, for example, a shape of the EA member 30 and/or a positional relationship between the EA member 30 and the battery pack 20.

Similarly, the position where the EA member 30 is fixed to the left rocker 24 of the vehicle body 12 does not need to be the second middle portion 38. The position where the EA member 30 is fixed to the left rocker 24 of the vehicle body 12 can suitably be changed depending on, for example, the shape of the EA member 30 and a positional relationship between the EA member 30 and the left rocker 24.

As illustrated in FIG. 3, the EA member 30 further includes a protrusion 35. The protrusion 35 is connected to the lower wall 34c and the inner sidewall 34a of the inner end portion 34 and extends inwardly in the vehicle width direction toward the support 20b disposed on the battery pack 20. The protrusion 35 is located below the battery pack 20 and extends along the lower wall 20d of the battery pack 20. The protrusion 35 is fixed to the lower wall 20d of the battery pack 20 with a third bolt 50. The tip end of the protrusion 35 comprises a contact surface 35a being in contact with a side surface 20c of the support 20b in the vehicle width direction. An entirety of the protrusion 35 needs not to be located below the battery pack 20, and only at least one of the protrusion 35 has to be located below the battery pack 20. As another embodiment, the protrusion 35 may be fixed to the lower wall 20d of the battery pack 20 with for example an adhesive, instead of the third bolt 50.

With the aforementioned structure, as illustrated in FIGS. 4A to 4C, one of the EA members 30, 32 deforms when side collision occurs to the vehicle 10, by which collision energy is absorbed. Each of the EA members 30, 32 includes the protrusion 35 which protrudes inwardly in the vehicle width direction toward the support 20b disposed on the lower wall 20d of the battery pack 20, and the tip end of the protrusion 35 comprises the contact surface 35a being in contact with the side surface 20c of the support 20b. Thus, when side collision occurs to the vehicle 10, collision load applied on one of the EA members 30, 32 is transmitted to the support 20b disposed on the lower wall 20d of the battery pack 20. Consequently, the collision load transmitted from the one of the EA members 30, 32 to the corresponding sidewall of the battery pack 20 can be reduced.

Although not particularly limited to this configuration, as illustrated in FIG. 5, the dimension H1 of the contact surface 35a of the protrusion 35 in the vehicle height direction is equal to or greater than the dimension H2 of the side surface 20c of the support 20b in the vehicle height direction, and more specifically, greater than the dimension H2 of the side surface 20c of the support 20b in the vehicle height direction. According to such a configuration, when side collision occurs to the vehicle 10, it is possible to avoid or suppress the protrusion 35 of one of the EA members 30, 32 being locally pressed against the corresponding side surface 20c of the support 20b disposed on the battery pack 20. Consequently, it is possible to avoid or suppress collision load transmitted to the support 20b being locally concentrated. As another embodiment, however, the dimension H1 of the contact surface 35a of the protrusion 35 in the vehicle height direction may be smaller than the dimension H2 of the side surface 20c of the support 20b in the vehicle height direction.

Although not particularly limited to this configuration, as illustrated in FIG. 5, the protrusion 35 has a plate-like shape extending in the vehicle width direction. In this case, the thickness T1 of the protrusion 35 is greater than the thickness of any one of the walls that constitute the rest of the EA member 30 (i.e., each of the thicknesses T2, T3 of the walls 34a to 34c that constitute the inner end portion 34, a thickness of any one of walls constituting the first middle portion 36, a thickness of each of the walls 38a, 38b that

constitute the second middle portion **38** and a thickness of any one of walls that constitute the outer end portion **40**). According to such a configuration, in the EA members **30**, **32**, it is possible to make rigidity of the protrusions **35** having the plate-like shape greater than rigidity of any one of the walls that constitute the rest of the EA members **30**, **32** except the protrusions **35**. Consequently, when one of the EA members **30**, **32** is compressed and deformed in the vehicle width direction toward the battery pack **20**, it is possible to make the corresponding protrusion **35** less susceptible to deformation as compared to any one of the walls that constitute the rest of the one of the EA members **30**, **32**, which is the inner end portion **34**, the first middle portion **36**, the second middle portion **38**, and the outer end portion **40**. Thus, the collision energy generated in side collision occurring to the vehicle **10** is primarily absorbed by deformation of a part of or the entirety of the rest of the one of the EA members **30**, **32** except the protrusion **35**.

Although not particularly limited to this configuration, as illustrated in FIG. **5**, in at least a part of the protrusion **35** adjacent to the contact surface **35a**, the thickness of the protrusion **35** increases toward the contact surface **35a**. In other words, the thickness **T1** of the protrusion **35** changes such that it increases toward the contact surface **35a**. For example, a thickness of a part of the protrusion **35** at which the third bolt **50** is disposed can relatively be reduced and a thickness of the contact surface **35a** at the tip end of the protrusion **35** (i.e., the dimension **H1** of the contact surface **35a** in the vehicle height direction) can relatively be increased. According to such a configuration, it is possible to make the minimum ground clearance of the protrusion **35** (i.e., a distance from the ground surface to the lowest surface of the protrusion **35** in the vehicle height direction) relatively great while rigidity of the contact surface **35a** of the protrusion **35** is maintained.

Alternatively to the aforementioned embodiment, as illustrated in FIGS. **6** to **9**, the thickness **T1** of the protrusion **35** may be constant. In this case, the thickness **T1** of the protrusion **35** may be equal to the dimension **H1** of the contact surface **35a** of the protrusion **35** in the vehicle height direction. The contact surface **35a** disposed at the tip end of the protrusion **35** and the side surface **20c** of the support **20b** can have various shapes. For example, as illustrated in FIGS. **7-9**, the contact surface **35a** disposed at the tip end of the protrusion **35** has a shape protruding inwardly in the vehicle width direction toward the side surface **20c** of the support **20b**. The side surface **20c** of the support **20b** is recessed inwardly in the vehicle width direction to receive the contact surface **35a** of the protrusion **35**. As described above, when the contact surface **35a** of the protrusion **35** and the side surface **20c** of the support **20b** have uneven shapes which engage each other, it is possible to increase an area where the contact surface **35a** of the protrusion **35** is in contact with the side surface **20c** of the support **20b**. In particular, when the uneven shapes are defined along the vehicle height direction, it is possible to suppress the contact surface **35a** of the protrusion **35** moving away from the side surface **20c** of the support **20b** upon side collision to the vehicle **10**. Consequently, when side collision occurs to the vehicle **10**, collision load applied on one of the EA members **30**, **32** is effectively transmitted to the support **20b** disposed on the lower wall **20d** of the battery pack **20** via the corresponding protrusion **35**.

What is claimed is:

1. A vehicle comprising:

a vehicle body comprising a floor panel;
a battery pack located below the floor panel; and
an energy absorption member located outwardly from the battery pack in a vehicle width direction and fixing the battery pack to the vehicle body,

wherein:

a support that protrudes downwardly is disposed on a lower wall of the battery pack,

the energy absorption member comprises a protrusion that protrudes inwardly in the vehicle width direction toward the support,

a tip end of the protrusion comprises a contact surface being in contact with a side surface of the support in the vehicle width direction, and

a dimension of the contact surface of the protrusion in a vehicle height direction is equal to or greater than a dimension of the side surface of the support in the vehicle height direction.

2. The vehicle according to claim **1**, wherein at least a part of the protrusion is located below the battery pack and extends along the lower wall of the battery pack.

3. The vehicle according to claim **2**, wherein the protrusion is fixed to the lower wall of the battery pack with a bolt.

4. The vehicle according to claim **1**, wherein, in at least a part of the protrusion adjacent to the contact surface, a thickness of the protrusion increases toward the contact surface.

5. The vehicle according to claim **1**, wherein the support is formed on a member that is attached to the battery pack.

6. The vehicle according to claim **1**, wherein the support is formed integrally with the battery pack.

7. The vehicle according to claim **1**, wherein the contact surface of the protrusion and the side surface of the support have uneven shapes that engage each other.

8. A vehicle comprising:

a vehicle body comprising a floor panel;
a battery pack located below the floor panel; and
an energy absorption member located outwardly from the battery pack in a vehicle width direction and fixing the battery pack to the vehicle body,

wherein:

a support that protrudes downwardly is disposed on a lower wall of the battery pack,

the energy absorption member comprises a protrusion that protrudes inwardly in the vehicle width direction toward the support,

a tip end of the protrusion comprises a contact surface being in contact with a side surface of the support in the vehicle width direction, and

a dimension of the contact surface of the protrusion in a vehicle height direction is greater than a dimension of the side surface of the support in the vehicle height direction.

9. A vehicle comprising:

a vehicle body comprising a floor panel;
a battery pack located below the floor panel; and
an energy absorption member located outwardly from the battery pack in a vehicle width direction and fixing the battery pack to the vehicle body,

wherein:

a support that protrudes downwardly is disposed on a lower wall of the battery pack, the energy absorption member comprises a protrusion that protrudes inwardly in the vehicle width direction toward the support,

11

a tip end of the protrusion comprises a contact surface being in contact with a side surface of the support in the vehicle width direction, and

at least a part of the protrusion is located below the battery pack and extends along the lower wall of the battery pack.

10. The vehicle according to claim 9, wherein the protrusion is fixed to the lower wall of the battery pack with a bolt.

* * * * *

10

12